

Environmental Pollution

For normal and healthy living a conducive environment is required by all living beings, including humans, livestock, plants, micro-organisms and the wildlife. The favourable unpolluted environment has a specific composition. When this composition gets changed by addition of harmful substances, the environment is called polluted environment and the substances polluting it are called pollutants. **Environmental pollution can, therefore, be defined as any undesirable change in the physical, chemical or biological characteristics of any component of the environment (air, water, soil), which can cause harmful effects on various forms of life or property.** Environmental pollution could be of various types:

6.1 WATER POLLUTION

Water pollution can be defined as alteration in physical, chemical or biological characteristics of water making it unsuitable for designated use in its natural state.

(A) Sources of Water Pollution: Water is an essential commodity for survival. We need water for drinking, cooking, bathing, washing, irrigation, and for industrial operations. Most of the water for such uses comes from rivers, lakes or groundwater sources. Water has the property to dissolve many substances in it, therefore, it can easily get polluted. Pollution of water can be caused by “point sources” or “non-point sources”. Point sources are specific sites near water which directly discharge effluents into them. Major point sources of water pollution are industries, power plants, underground coal mines, offshore oil wells etc. The discharge from non-point sources is not at any particular site, rather, these sources are scattered, which individually or collectively pollute water. Surface run-off from agricultural fields, overflowing small drains, rain water sweeping roads and fields, atmospheric deposition etc., are the non-point sources of water pollution.

Ground Water Pollution: Ground water forms about 0.6% of the total water available on planet earth and is about 30 times more than surface water (streams, lakes and estuaries). Ground water seems to be less prone to pollution as the soil mantle through which water passes helps to retain various contaminants due to its cation exchange capacity. However, there are a number of potential sources of ground water pollution. Septic tanks, industry (textile, chemical, tanneries), deep well injection, mining

etc., are mainly responsible for ground water pollution, which is irreversible. Ground water pollution with arsenic, fluoride and nitrate are posing serious health hazards.

Surface water pollution: The major sources of surface water pollution are:

1. **Sewage.** Emptying the drains and sewers in fresh water bodies causes water pollution. The problem is severe in cities.
2. **Industrial effluents.** Industrial wastes containing toxic chemicals, acids, alkalis, metallic salts, phenols, cyanides, ammonia, radioactive substances, etc., are sources of water pollution. They also cause thermal (heat) pollution of water.
3. **Synthetic detergents.** Synthetic detergents used in washing and cleaning produce foam and pollute water.
4. **Agrochemicals.** Agrochemicals like fertilizers (containing nitrates and phosphates) and pesticides (insecticides, fungicides, herbicides etc.) washed by rain-water and surface run-off pollute water.
5. **Oil.** Oil spillage into sea-water during drilling and shipment pollute it.
6. **Waste heat.** Waste heat from industrial discharges increases the temperature of water bodies and affects distribution and survival of sensitive species.

(B) Effects of Water Pollution: Following are some important effects of various types of water pollutants:

Oxygen demanding wastes: Organic matter which reaches water bodies is decomposed by micro-organisms present in water. For this degradation, oxygen dissolved in water is consumed. Dissolved oxygen (DO) is the amount of oxygen dissolved in a given quantity of water at a particular temperature and atmospheric pressure. Amount of dissolved oxygen depends on aeration, photosynthetic activity in water, respiration of animals and plants and ambient temperature.

The saturation value of DO varies from 8–15 mg/L. For active fish species (trout and Salmon) 5–8 mg/L of DO is required whereas less desirable species like carp can survive at 3.0 mg/L of DO.

Lower DO may be harmful to animals especially fish population. Oxygen depletion (deoxygenation) helps in release of phosphates from bottom sediments and causes eutrophication.

Nitrogen and phosphorus compounds (nutrients): Addition of compounds containing nitrogen and phosphorus helps in the growth of algae and other plants which when die and decay consume oxygen of water. Under anaerobic conditions foul smelling gases are produced. Excess growth or decomposition of plant material will change the concentration of CO_2 which will further change pH of water. Changes in pH, oxygen and temperature will change many physico-chemical characteristics of water.

Pathogens: Many waste waters especially sewage contain many pathogenic (disease causing) and non-pathogenic micro-organisms and many viruses. Water borne diseases like cholera, dysentery, typhoid, jaundice etc. are spread by water contaminated with sewage. Detailed account has been given in chapter IV.

Toxic compounds: Pollutants such as heavy metals, pesticides, cyanides and many other organic and inorganic compounds are harmful to aquatic organisms.

The demand of DO increases with addition of biodegradable organic matter which is expressed as biological oxygen demand (BOD). BOD is defined as the amount of DO required to aerobically decompose biodegradable organic matter of a given volume of water over a period of 5 days at 20°C. Higher BOD values of any water sample are associated with poor water quality. The non-biodegradable toxic compounds biomagnify in the food chain and cause toxic effects at various levels of food chain.

Some of these substances like pesticides, methyl mercury etc., move into the bodies of organisms from the medium in which these organisms live. Substances like DDT are not water soluble and have affinity for body lipids. These substances tend to accumulate in the organism's body from the medium food. This process is called **bioaccumulation** or **bioconcentration**. The concentration of these toxic substances builds up at successive levels of food chain. This process is called **biomagnification**. Following is the example of biomagnification of DDT in aquatic food chain:

Component	DDT concentration (ppm)
Birds	10.00
↑	↑
Needle fish	1.0
↑	↑
Minnows	0.1
↑	↑
Zooplankton	0.01
↑	↑
Water	0.000001

Toxic substances polluting the water ultimately affect human health. Some heavy metals like lead, mercury and cadmium cause various types of diseases. Mercury dumped into water is transformed into water soluble methyl mercury by bacterial action. Methyl mercury accumulates in fish. In 1953, people in Japan suffered from numbness of body parts, vision and hearing problems and abnormal mental behaviour. This disease called **Minamata disease** occurred due to consumption of methyl mercury contaminated fish caught from Minamata Bay in Japan. The disease claimed 50 lives and permanently paralysed over 700 persons. Pollution by another heavy metal cadmium had caused the disease called **Itai-itai** in the people of Japan. The disease was caused by cadmium contaminated rice. The rice fields were irrigated with effluents of zinc smelters and drainage water from mines. In this disease bones, liver, kidney, lungs, pancreas and thyroid are affected.

Lead in water may be released from water pipes as lead is used in plumbing. Lead poisoning affects kidneys reproductive system, liver, brain and central nervous system. It also causes anemia and mental retardation in children.

Arsenic Pollution of Ground Water

Arsenic Pollution in Groundwater: West Bengal and Bangladesh are severely contaminated by the toxic heavy metal arsenic. The first report of arsenic pollution in West Bengal came in 1978 and that of Bangladesh in 1993, where it was found to be even more widespread. Arsenic poisoning has far reaching consequences. The local people were found to be ingesting low doses of arsenic for 10–14 years after which suddenly white or black spots called melanosis started mottling the skin. The spots were later found to get converted into leprosy like skin lesions encrusting the palms and soles, eventually rotting into gangrenous ulcers. Long exposures often led to bladder and lung cancer. Children are more badly affected by arsenicosis, the affected people are socially isolated, children barred from attending schools and young women remain single or have broken marriages. The WHO has prescribed the maximum permissible limits of arsenic as 10 mg/L. In West Bengal 40 million out of 90 million people are feared to have likely exposure to arsenic threat due to contaminated water. The 24 Paraganas, Hooghly and Murshidabad districts as also Behala and South Eastern fringes of Kolkata lie in Arsenic Risk Zone. Earlier it was postulated that arsenic has entered into groundwater due to geologic reasons in the Ganga Delta. Recently, however, it is being linked with anthropogenic causes.

Excessive use of lead arsenate and copper arsenite as pesticides in high yielding varieties of summer paddy and jute crop seems to be the major cause of arsenic pollution. Now the arsenic contaminated tubewells in the state are being painted red while safe water tubewells are painted green for use by people.

Nitrate ions present in the water are harmful to human health. From nitrogen fertilizers, nitrate ions seep into water bodies from where these may bioaccumulate in the bodies of the consumers. In the stomach nitrate is reduced to nitrite and is responsible for **blue baby syndrome** and stomach cancer. Young children less than 1 year of age when ingest excessive nitrate, nitrite is formed in their stomach by microbes as the stomach is not acidic to the extent to inhibit the microbes. Which bring about this conversions. Nitrite reacts with haemoglobin which is converted into non-functional oxidised form. Due to diarrhoea and vomiting child becomes slate blue, giving the baby blue baby syndrome or **methaemoglobinaemia**.

Nitrate in the stomach of adult humans partly converts into nitrite. Nitrite interacts with secondary amines to produce N-nitrosamine which in experimental animals have been shown to cause stomach cancer. In human beings, however, evidences do not show association of nitrate with cancer.

Fluoride pollution causes defects in teeth and bones, a disease called **fluorosis**. Details have been given in chapter 4. Pesticides through drinking water reach humans and are known to cause various health problems. The organophosphorus and carbamate pesticides are more toxic. Organochlorine pesticides accumulate in the body and affect various organs especially the central nervous system. These pesticides stimulate liver enzymes which results in rapid metabolism of drugs in the person who is on medication. The effectiveness of medicines will be reduced.

Since some pesticide accumulate in the body, these may be transferred to young babies with the mothers's milk. Due to their persistence DDT, aldrin, dieldrin etc. have been banned in many countries. Some years ago people in Andhra Pradesh suffered from various abnormalities due to consumption of endosulphan contaminated cashew nuts.

(C) Remedial Measures for Water Pollution (Control): It is easy to reduce water pollution from point sources by legislation. However, due to absence of defined strategies it becomes difficult to prevent water pollution from non-point sources. The following points may help in reducing water pollution from non-point sources.

(i) Judicious use of agrochemicals like pesticides and fertilizers which will reduce their surface run-off and leaching. Use of these on sloped lands should be avoided.

(ii) Use of nitrogen fixing plants to supplement the use of fertilizers.

(iii) Adopting integrated pest management to reduce greater reliance on pesticides.

(iv) Prevent run-off of manure. Divert such run-off to basin for settlement. The nutrient rich water can be used as fertilizer in the fields.

(v) Separate drainage of sewage and rain water should be provided to prevent overflow of sewage with rain water.

(vi) Planting trees would reduce pollution by sediments and will also prevent soil erosion.

For controlling water pollution from point sources, treatment of waste waters is essential before being discharged. Parametres which are considered for reduction in such water are:

Total solids, biological oxygen demand (BOD), chemical oxygen demand (COD), nitrates and phosphates, oil and grease, toxic metals etc.

Waste waters should be properly treated by primary and secondary treatments to reduce the BOD, COD levels upto the permissible levels for discharge.

6.2 LAND POLLUTION

(A) Sources of Land Pollution: Soil is the upper layer of the earth crust which is formed by weathering of rocks. Organic matter in the soil makes it suitable for living organisms. Dumping of various types of materials especially domestic and industrial wastes causes land pollution. Domestic wastes include garbages, rubbish material like glass, plastics, polythene bags, metallic cans, paper, fibres, cloth rags, containers, paints, varnishes etc. Leachates from dumping sites and sewage tanks are harmful and toxic, which pollute the soil. Polythene does not degrade and becomes brittle in due course. It affects the porosity of the soil.

Industrial wastes are the effluents discharged from chemical industries, paper and pulp mills, tanneries, textile mills, steel industries, distilleries, refineries, pesticides and fertilizer industries, pharmaceutical industries, food processing industries, cement industries, thermal and nuclear power plants, mining industries etc. Thermal power plants generate a large quantity of 'Fly ash'. Huge quantities of these wastes are dumped on land which cause land pollution.

Pesticides are used to kill pests that damage crops. These pesticides ultimately reach the soil and persist there for a long time. Pesticides which are persistent in nature are chlorinated hydrocarbon insecticides *e.g.*, DDT, HCH, endrin, lindane, heptachlor, endosulfan etc. Residues of these pesticides in soils have long term effects especially under the temperate conditions.

Industrial wastes also contain some organic and inorganic compounds that are refractory and non-biodegradable. Industrial sludge may contain various salts, toxic substances, metals like mercury, lead, cadmium, arsenic etc. Agrochemicals released with the wastes of pesticide and fertilizer factories or during agricultural practices also reach land and pollute it.

Land also receives excreta from animals and humans. The sewage sludge contains many pathogenic organisms, bacteria, viruses and intestinal worms which cause pollution of the land.

The sources of radioactive substances in soil are explosion of radioactive devices, radioactive wastes discharged from industries and laboratories, aerial fall-out etc. Isotopes of radium, uranium, thorium, strontium, iodine, caesium and of many other elements reach land and persist there for a long time and keep on emitting radiations.

(B) Effects of Land Pollution: Sewage and industrial effluents which pollute land ultimately affect human health. Various types of chemicals like acids, alkalis, pesticides, insecticides, weedicides, fungicides, heavy metals etc., in the industrial discharges affect soil fertility by causing changes in physical, chemical and biological properties.

Some of the persistent toxic chemicals inhibit the non-target organisms, soil flora and fauna and reduce soil productivity. These chemicals accumulate in food chain and ultimately affect human health. Indiscriminate use of pesticides specially is a matter of concern.

Sewage sludge has many types of pathogenic bacteria, viruses and intestinal worms which may cause various types of diseases. Decomposing organic matter in soil also produces toxic vapours.

Radioactive fallout on vegetation is the source of radio-isotopes which enter the food chain in the grazing animals. Some of these radio-isotopes replace essential elements in the body and cause abnormalities *e.g.* strontium-90 instead of calcium gets deposited in the bones and tissues. The bones become brittle and prone to fracture.

Radio-isotopes which attach with the clay become a source of radiations in the environment.

Nitrogen and phosphorus from the fertilizers in land reach nearby water bodies with agricultural run-off and cause **eutrophication**. Chemicals or their degradation product from land may percolate and contaminate ground water resources.

(C) Control of Land Pollution

- (i) Effluents should be properly treated before discharging them on land.
- (ii) Solid wastes should be properly collected and disposed off by appropriate method.
- (iii) From the wastes, recovery of useful products should be done.
- (iv) Biodegradable organic waste should be used for generation of biogas.

- (v) Cattle dung should be used for methane generation. Night-soil (human faeces) can also be used in the biogas plant to produce inflammable methane gas.
- (vi) Microbial degradation of biodegradable substances is also one of the scientific approaches for reducing land pollution.

6.3 NOISE POLLUTION

We hear various types of sounds everyday. Sound is mechanical energy from a vibrating source. A type of sound may be pleasant to someone and at the same time unpleasant to others. The unpleasant and unwanted sound is called noise.

Sound can propagate through a medium like air, liquid or solid. Sound wave is a pressure perturbation in the medium through which sound travels. Sound pressure alternately causes compression and rarefaction. The number of compressions and rarefaction of the molecules of the medium (for example, air) in a unit time is described as frequency. It is expressed in Hertz (Hz) and is equal to the number of cycles per second.

There is a wide range of sound pressures, which encounter human ear. Increase in sound pressure does not invoke linear response of human ear. A meaningful logarithmic scale has been devised. The noise measurements are expressed as Sound Pressure Level (SPL) which is logarithmic ratio of the sound pressure to a reference pressure. It is expressed as a dimensionless unit, decibel (dB). The international reference pressure of 2×10^{-5} Pa is the average threshold of hearing for a healthy ear. Decibel scale is a measure of loudness. Noise can affect human ear because of its loudness and frequency (pitch).

The Central Pollution Control Board (CPCB) has recommended permissible noise levels for different locations as given in Table 6.1.

Table 6.1. Noise Standards Recommended by CPCB Committee

Area code	Category of area	Noise level in dB(A) <i>Leq</i>	
		Day	Night
(A)	Industrial	75	70
(B)	Commercial	65	55
(C)	Residential	55	45
(D)	Silence zone	50	40

Table 6.2. Different Sounds and their Sound Levels on Decibel Scale

Sound level (dB)	Source of sound
180 —	Rocket engine
170	
160	
150 —	Jet plane take off
Threshold of pain — 140	
130 —	Maximum recorded rock music
120 —	Thunder cap
110 —	Autohorn 1 m away
100 —	Jet fly over at 300 m, construction work, newspaper press
90 —	Motor cycle/8 m away, food blender
80	
70 —	Vacuum cleaner, ordinary conversation
60 —	Air conditioning unit, 6 m away; light traffic noise, 30 m away
50 —	Average living room
40	
30 —	Library, soft whisper
20 —	Broadcasting studio
10 —	Rustling leaf
Threshold of hearing — 0	

(A) Sources of Noise Pollution: The sources of noise can be classified as (i) Mobile sources and (ii) Stationary sources. (i) Mobile sources are various modes of transportation (like air, road, rail-transportation) and (ii) Stationary sources include industrial operations, construction activities and celebrations (social/religious functions, elections etc.), electric home appliances etc.

High levels of noise have been recorded in some of the cities of the world. In Nanjing (China) noise level of 105 dB has been recorded, while in some other cities of the world these levels are: Rome 90 dB, New York 88 dB, Kolkata 85 dB, Mumbai 82 dB, Delhi 80 dB, Kathmandu 75 dB.

(B) Effects of Noise Pollution: Noise pollution causes the following effects:

(i) Interferes with man's communication: In a noisy area communication is severely affected.

(ii) Hearing damage: Noise can cause temporary or permanent hearing loss. It depends on intensity and duration of sound level. Auditory sensitivity is reduced with noise level for over 90 dB in the mid-high frequency for more than a few minutes.

(iii) Physiological and psychological changes: Continuous exposure to noise affects the functioning of various systems of the body. It may result in hypertension, insomnia (sleeplessness), gastro-intestinal and digestive disorders, peptic ulcers, blood pressure changes, behavioural changes, emotional changes etc.

(C) Noise Pollution During Diwali: Diwali is a festival of lights. Traditionally people of all ages enjoy firecrackers. Some accidents do occur every year claiming a few lives. Besides, noise generated by various firecrackers is beyond the permissible noise levels of 125 decibels as per the Environmental (Protection) (Second Amendment) Rules, 1999.

There has been a great concern over the noise levels generated during Diwali. Some measurements by certain group of researchers have also been made at various places during Diwali. It is recommended that the manufacturers of fireworks should mention the noise levels in decibels generated by individual items. The department of explosives of the Union Ministry of Commerce and Industry is entrusted with the task to ensure that the industry produces firecrackers conforming to permissible noise standards.

According to a test report on firecrackers produced by the National Physical Laboratory, New Delhi most of the firecrackers available in the market produce noise beyond the permissible levels of 125 decibels as per the Environment (Protection) (Second amendment) Rules, 1999. Some of them have been observed to produce noise near the threshold of pain. The details are given in Table 6.3.

Table 6.3. Noise Levels Generated by Firecrackers

<i>Types of firecracker</i>	<i>Manufacturer</i>	<i>Generated noise level in decibels</i>
Atom bomb (timing bomb)	Coronation Fireworks, Sivakasi	135 $\pm$ 2
Chinese crackers (a string of 1,000 in one piece)	Sri Kaliswari Fireworks, Sivakasi	128
Chinese crackers (a string of 600 in one piece)	Sri Kaliswari Fireworks, Sivakasi	132
Nazi (atom bomb)	Coronation Fireworks, Sivakasi	135 $\pm$ 0
Magic formula (flower bomb)	Rajan Fireworks, Sivakasi	136 $\pm$ 1
Atom bomb (foiled)	Sri Kaliswari Fireworks, Sivakasi	131 $\pm$ 2
Hydrogen bomb	Sri Patrakali Fireworks, Sivakasi	134 $\pm$ 2
Rajan classic dhamaka (foiled bomb)	Rajan Fireworks, Sivakasi	136 $\pm$ 0
Samrat classic bomb (deluxe)	Venkateswara Fireworks, Sivakasi	136 $\pm$ 0
Hydro foiled (bomb)	Sri Kaliswari Fireworks, Sivakasi	132 $\pm$ 2
*Three sound (bomb)	Coronation Fireworks, Sivakasi	119 $\pm$ 7
Atom bomb	Local	136 $\pm$ 0

*Cracker meeting the noise pollution standards.

Source: Test report on firecrackers, National Physical Laboratory, New Delhi, April 21, 2003.

The noise levels were measured under standard conditions *i.e.* in areas not having noise-reflecting surfaces within a 15 metre radius. Two gadgets, for measuring sound levels were installed at a height of 1.3 metres and at a distance of 4 metres from the source of sound.

Besides mentioning the sound levels on each of the types of firecrackers or banning the production of such firecrackers which produce noise above permissible levels, it is important to educate people about the harmful effects of noise during such festivals like Diwali. It can be done by giving public notices in the leading newspapers and messages through other mass media like radio and television.

Honourable Supreme Court in a Writ Petition (civil) of 1998 concerning noise pollution had passed the following directions as an interim measure.

The Union Government, The Union Territories as well as all the State Governments shall in particular comply with amended Rule 89 of the Environmental (Protection) Rules, 1986 framed under the Environmental (Protection) Act, 1986 which essentially reads as follows:

1. (i) The manufacture, sale or use of firecrackers generating noise level exceeding 125 dB (AI) or 145 dB (C) pk at 4 meters distance from the point of bursting shall be prohibited.
(ii) For individual firecracker constituting the series (joined firecrackers), the above mentioned limit be reduced by $5 \log_{10} (N)$ dB, where N = Number of crackers joined together.
2. The use of fireworks or firecrackers shall not be permitted except between 6.00 p.m. and 10.00 p.m. No fireworks or firecrackers shall be used between 10.00 p.m. and 6.00 a.m.
3. Firecrackers shall not be used at any time in silence zones, as defined by the Ministry of Environment and Forests. Silence Zone has been defined as:
“Silence Zone in an area comprising not less than 100 meters around hospitals, educational institutions, courts, religious places or any other area which is declared as such by the competent authority.”
4. The State Education Resource Centres in all the States and Union Territories as well as the management/principals of schools in all the States and Union Territories shall take appropriate steps to educate students about the ill effects of air and noise pollution and apprise them of directions (1) to (3) above.

(D) Remedial Measures for Noise Pollution (Control)

1. *Reduction in sources of noise:* Sources of noise pollution like heavy vehicles and old vehicles may not be allowed to ply in the populated areas.
2. Noise making machines should be kept in containers with sound absorbing media. The noise path will be uninterrupted and will not reach the workers.
3. Proper oiling will reduce the noise from the machinery.
4. *Use of sound absorbing silencers:* Silencers can reduce noise by absorbing sound. For this purpose various types of fibrous material could be used.
5. Planting more trees having broad leaves.
6. *Through law:* Legislation can ensure that sound production is minimised at various social functions. Unnecessary horn blowing should be restricted especially in vehicle-congested areas.

6.4 PUBLIC HEALTH ASPECTS

Various aspects concerning protection and promotion of healthy human environment including water, air, food, shelter and working areas are known as public health aspects. For promotion of public health, diverse knowledge inputs from various fields like ecology, microbiology, pathology, entomology, epidemiology, human physiology and engineering are required. Public health basically deals with water supply, sewerage and sewage disposal, drainage system, refuse sanitation, ventilation, air conditioning, and air pollution abatement and prevention of outbreak of diseases like malaria, dengue, encephalitis etc. The engineers of public health department apply engineering principles for sanitation of the environment.

6.4.1 Public Water Supply

All our public water supplies should be free from various types of impurities and the water should meet the drinking water quality standards prescribed by agencies like Bureau of Indian Standards (BIS), World Health Organization (WHO) or Indian Council of Medical Research (ICMR).

In ancient times mainly ground water was used that was derived from wells or springs. Since groundwater moves through porous strata of sediments of the soil, it was largely filtered and pure. When surface waters were to be used, these were made to stay in impounding reservoirs so that the impurities settled down. Then, **filtration** through beds of sand and gravel was used followed by flocculation using alum. This process was known as **coagulation**. However, the filtered and coagulated water is still not free of microbes responsible for spreading various diseases like cholera, typhoid, dysentery etc. Therefore, **disinfection** of water becomes a very important step in public water supply systems.

Some important methods of disinfection are:

- (i) **Chlorination.** Chlorine or chlorinous compounds like bleaching powder [$\text{Ca}(\text{OCl})_2$] or sodium hypochlorite (Na OCl) are used for chlorination. The dosage of chlorine has to be determined first, because too little chlorine is ineffective, while too much may cause bad taste and odour and objectionable complexes.
- (ii) **Ozonation.** Ozone (O_3), while changing into its stable molecular form (O_2) releases nascent oxygen (O). The high oxidizing power of ozone destroys bacterial spores, without causing bad taste or odour.
- (iii) **UV-rays.** Ultraviolet (UV) rays are powerful disinfectant. The rays are generated by passing electric current through mercury vapour lamp enclosed in quartz bulb. Water is passed over the lamp. Though it causes no objectionable taste and odour, but has limited application due to high cost.
- (iv) **Iodine and bromine.** These are quite cheap, and have disinfecting power. However, they give a clinical taste to water.
- (v) **Potassium permanganate.** This is an effective disinfectant due to its oxidizing property and has been found to be effective against cholera only.
- (vi) **Copper sulphate treatment.** It is mainly used to control algal growth in water supply system.

- (vii) **Activated carbon.** It is the most important method for control of taste and odour. It is very porous and can remove majority of the dissolved impurities in water. It also acts as an agent to remove excess chlorine, phenolic taste and colour from water.

To make water easily accessible, a well laid-out system of water distribution involving pipes, valves and other fixtures of appropriate design and capacity are required.

6.4.2 Sewerage and Sewage Disposal

The wastewater originating from community usage including domestic, industrial and storm water is termed as **sewage**. The underground conduit used for carrying and removing the sewage is known as **sewerage**.

The sewage is split up solid, liquid and gaseous components. The semi solid part (sludge) is dried, buried or dumped, or may be used as a fertilizer. The liquid part is used in ferti-irrigation projects. The gases (like methane) may be used as an energy source.

The sewer systems are of different types:

(i) **Separate System.** The domestic sewage and industrial wastewater are carried in one set of sewers, whereas storm water and surface water are in another set.

(ii) **Combined System:** All the above types of wastewaters are carried through the same set. Sometimes partially separate systems are used, where domestic sewage, industrial sewage and also a part of surface water drained from roof tops and streets are carried through same sewer.

A proper sewerage system is designed keeping into consideration the population (present and future), type of area served (population density, land-use pattern), rate of water supply and ground water infiltration (depending upon nature of soil, depth of sewer, types of material used etc.).

The fresh sewage gives obnoxious smell and is a breeding ground for pathogens. Therefore, public health department takes care for proper decomposition of the sewage.

First aerobic bacteria are made to break down the complex organic matter in the presence of some oxygen. This is followed by **putrefaction**, when anaerobic bacteria break down the organic compounds, releasing CO_2 , H_2S , NH_3 and CH_4 . Once again aerobic bacteria carry out oxidation leading to formation of stable compounds with little odour. Septic tanks, Inhoff tanks and sludge digestion tanks carry out putrefaction, whereas contact beds, trickling filters and aeration tanks are meant for oxidation.

6.4.3 Drainage System

A proper drainage system is required to quickly remove all foul and waste matter from various fixtures. Some important considerations are:

- (i) The passage of gases and odour from the sewer into the building must be prevented.
- (ii) The drainage pipes should be strong, durable, corrosion resistant, air-tight and gas-tight.
- (iii) The pipe joints should be leak-proof.
- (iv) There should be ample means for cleaning and removal of obstruction from entire network of pipes.

6.4.4 Environmental Sanitation

It involves a system where the solid organic waste (refuse) is properly disposed off or converted into compost. Disposal may be done in dump-sites or sanitary landfills.

In houses or working places, proper ventilation and air conditioning is done to control temperature, humidity movement of air and clean and comfortable atmosphere. Another important environmental sanitation aspect is the prevention of outbreak of diseases. Spraying or fogging is done to kill the adult or larvae of mosquitoes or other vectors (carriers) of diseases like malaria, dengue etc. Animal wastes is also utilized for generation of biogas in villages and towns in big biogas plants.

6.5 AIR POLLUTION

It is an atmospheric condition in which certain substances (including the normal constituents in excess) are present in such concentrations which can cause undesirable effects on man and his environment. These substances include gases, particulate matter, radioactive substances etc.

Gaseous pollutants include oxides of sulphur (mostly, SO_2 , SO_3) oxides of nitrogen (mostly, NO and NO_2 or NO_x), carbon monoxide (CO), volatile organic compounds (mostly, hydrocarbons) etc. Particulate pollutants include smoke, dust, soot, fumes, aerosols, liquid droplets, pollen grains etc.

Radioactive pollutants include radon-222, iodine-131, strontium-90, plutonium-239 etc.

Classification of Air Pollutants

Air pollutants may occur in gaseous or particulate form and may be organic or inorganic in nature. On the basis of origin of pollutants they can be classified as primary or secondary pollutants.

Primary pollutants: These are emitted directly from the point source (identifiable source) *e.g.* carbon monoxide (CO), oxides of nitrogen (NO_x), oxides of sulphur (SO_x), hydrocarbons, radioactive substances etc.

Secondary pollutants: These are formed by interaction of primary pollutant(s) with other primary pollutant(s) or with some natural constituents of the atmosphere, *e.g.* ozone (O_3), peroxyacetyl nitrate (PAN), photochemical smog etc.

(A) Causes/Sources of Air Pollutants: The sources of air pollution are natural and man-made (anthropogenic).

- **Natural Sources:** The natural sources of air pollution are volcanic eruptions, forest fires, sea salt sprays, biological decay, photochemical oxidation of terpenes, marshes, extra terrestrial bodies, pollen grains of flowers, spores etc. Radioactive minerals present in the earth crust are the sources of radioactivity in the atmosphere.
- **Man-made:** Man-made sources include thermal power plants, industrial units, vehicular emissions, fossil fuel burning, agricultural activities etc. Thermal power plants have become the major sources for generating electricity in India as the nuclear power plants couldn't be installed as planned. The main pollutants emitted are fly ash and SO_2 . Metallurgical plants also consume coal and produce similar pollutants. Fertilizer plants, smelters, textile mills, tanneries, refineries, chemical industries, paper and pulp mills are other sources of air pollution.

Automobile exhaust is another major source of air pollution. Automobiles release gases such as carbon monoxide (about 77%), oxides of nitrogen (about 8%) and hydrocarbons (about 14%). Heavy duty diesel vehicles spew more NO_x and suspended particulate matter (SPM) than petrol vehicles which produce more carbon monoxide and hydrocarbons.

Indoor air pollution: The most important indoor air pollutant is radon gas. Radon gas and its radioactive daughters are responsible for a large number of lung cancer deaths each year. Radon can be emitted from building materials like bricks, concrete, tiles etc., which are derived from soil containing radium. Radon is also present in groundwater and natural gas and is emitted indoors while using them.

Many houses in the under-developed and developing countries including India use fuels like coal, dung-cakes, wood and kerosene in their kitchens. Complete combustion of fuel produces carbon dioxide which may not be toxic. However, incomplete combustion produces the toxic gas carbon monoxide. Coal contains varying amounts of sulphur which on burning produces sulphur dioxide. Fossil fuel burning produces black soot. These pollutants i.e. CO , SO_2 , soot and many others like formaldehyde, benzo-(a) pyrene (BAP) are toxic and harmful for health. BAP is also found in cigarette smoke and is considered to cause cancer. A housewife using wood as fuel for cooking inhales BAP equivalent to 20 packets of cigarette a day.

(B) Effects of Air Pollution: Air pollution has adverse effects on living organisms and materials.

- **Effects on human health.** Human respiratory system has a number of mechanisms for protection from air pollution. Bigger particles ($> 10 \mu\text{m}$) can be trapped by the hairs and sticky mucus in the lining of the nose. Smaller particles can reach tracheobronchial system and there get trapped in mucus. They are sent back to throat by beating of hair like cilia from where they can be removed by spitting or swallowing. Years of exposure to air pollutants (including cigarette smoke) adversely affects these natural defences and can result in lung cancer, asthma, chronic bronchitis and emphysema (damage to air sacs leading to loss of lung elasticity and acute shortness of breath). Suspended particulates can cause damage to lung tissues and diseases like asthma, bronchitis and cancer especially when they bring with them cancer causing or toxic pollutants attached on their surface. Sulphur dioxide (SO_2) causes constriction of respiratory passage and can cause bronchitis like conditions. In the presence of suspended particulates, SO_2 can form acid sulphate particles, which can go deep into the lungs and affect them severely.

Oxides of nitrogen especially NO_2 can irritate the lungs and cause conditions like chronic bronchitis and emphysema. Carbon monoxide (CO) reaches lungs and combines with haemoglobin of blood to form carboxyhaemoglobin. CO has affinity for haemoglobin 210 times more than oxygen. Haemoglobin is, therefore, unable to transport oxygen to various parts of the body. This causes suffocation. Long exposure to CO may cause dizziness, unconsciousness and even death.

Many other air pollutants like benzene (from unleaded petrol), formaldehyde and particulates like polychlorinated biphenyls (PCBs), toxic metals and dioxins (from burning of polythene) can cause mutations, reproductive problems or even cancer.

Many other hazardous materials like asbestos, beryllium, mercury, arsenic and radioactive substances cause lung diseases and/or affect other vital organs like kidney, liver, spleen, brain and some may also cause cancer.

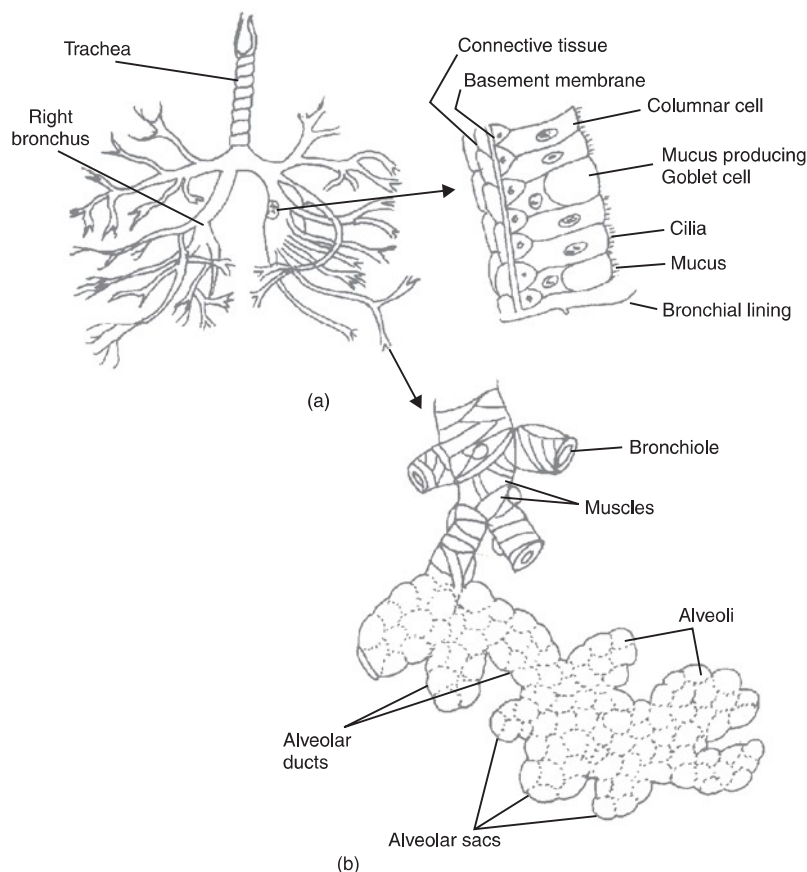


Fig. 6.1. Lower respiratory system of human beings (a and b) and cross-section of bronchial lining showing cilia and goblet cells.

- Effects on plants:** Air pollutants affect plants by entering through stomata (leaf pores through which gases diffuse), destroy chlorophyll and affect photosynthesis. During the day time the stomata are wide open to facilitate photosynthesis. Air pollutants during day time affect plants by entering the leaf through these stomata more than night. Pollutants also erode waxy coating of the leaves called cuticle. Cuticle prevents excessive water loss and damage from diseases, pests, drought and frost. Damage to leaf structure causes *necrosis* (dead areas of leaf), *chlorosis* (loss or reduction of chlorophyll causing yellowing of leaf) or *epinasty* (downward curling of leaf), and *abscission* (dropping of leaves). Particulates deposited on leaves can form encrustations and plug the stomata and also reduce the availability of sunlight. The damage can result in death of the plant.

SO₂ causes bleaching of leaves, chlorosis, injury and necrosis of leaves. NO₂ results in increased abscission and suppressed growth. O₃ causes flecks on leaf surface, premature aging, necrosis and bleaching. Peroxyacetyl nitrate (PAN) causes silverying of lower surface of leaf, damage to young and more sensitive leaves and suppressed growth. Fluorides cause necrosis of leaf-tip while ethylene results in epinasty, leaf abscission and dropping of flowers.

- **Effects on aquatic life:** Air pollutants mixing up with rain can cause high acidity (lower pH) in fresh water lakes. This affects aquatic life especially fish. Some of the fresh water lakes have experienced total fish death.
- **Effects on materials:** Because of their corrosiveness, particulates can cause damage to exposed surfaces. Presence of SO_2 and moisture can accelerate corrosion of metallic surfaces due to formation of sulfuric acid. Metal parts of buildings, vehicles, bridges, wires and metallic railway tracks are affected. Sulfuric acid also damages buildings and causes disfigurement of statues made up of marble and limestone. Sulfuric acid formed by the atmospheric SO_2 and water vapours damages the leather binding of books. The pages of the books also become brittle. SO_2 can affect fabric, leather, paint and paper. Ozone in the atmosphere can cause cracking of rubber. Nylon stockings are weakened and ultimately damaged. Tyres of various vehicles are also damaged. These days chemicals are added to prevent damage to tyre rubber by ozone. Oxides of nitrogen and ozone can also cause fading of cotton and rayon fibres.

(C) Remedial Measures for Air Pollution (Control): Air pollution can be minimised by the following methods:

- Siting of industries after proper environmental impact assessment studies.
- By dilution of emission. This can be done by increasing the stack height (though up to permissible height), beyond inversion layer. Wind currents will disperse the pollutants. But this results in interstate dispute and is not considered to be solution for air pollution problem.
- Minimise/modify activities which cause pollution *e.g.* transportation and energy production.
- Modification of process and/or equipment.
- Use of appropriate material.
- Using low sulphur coal in industries.
- Removing sulphur from coal (by washing or with the help of bacteria).
- Removing NO_x during the combustion process and controlling the flow of air and fuel in industrial boilers.
- Vehicular pollution can be checked by regular tune-up of engines; replacement of more polluting old vehicles; installing catalytic converters; by engine modification to have fuel efficient (lean) mixtures to reduce CO and hydrocarbon emissions; and slow and cooler burning of fuels to reduce NO_x emission (Honda Technology).
- Using mass transport system, bicycles etc.
- Shifting to less polluting (clean) fuels (hydrogen gas).
- Using non-conventional sources of energy.
- Using biological filters and bio-scrubbers.
- Planting more trees.
- Reduction of pollution at source by installing air pollution control devices like cyclones, baghouse filters, wet scrubbers, electrostatic precipitators etc.

6.6 SOLID WASTE MANAGEMENT

Higher standards of living of ever increasing population has resulted in an increase in the quantity and variety of waste generated. It is now realized that if waste generation continues indiscriminately then very soon it would be beyond rectification. Management of solid waste has, therefore, become very important in order to minimize the adverse effects of solid wastes. Solid waste (waste other than liquid or gaseous) can be classified as municipal, industrial, agricultural, medical, mining waste and sewage sludge.

(A) Sources of Urban and Industrial Wastes: Urban waste consists of medical waste from hospitals; municipal solid wastes from homes, offices, markets (commercial waste) small cottage units, and horticulture waste from parks, gardens, orchards etc.

- **Waste from homes (Domestic waste)** contains a variety of discarded materials like polyethylene bags, empty metal and aluminium cans, scrap metals, glass bottles, waste paper, diapers, cloth/rags, food waste, electronic waste (e-waste).
- **Waste from shops** mainly consists of waste paper, packaging material, cans, bottles, polyethylene bags, peanut shells, egg shells, tea leaves etc.
- **Biomedical waste** includes anatomical wastes, pathological wastes, infectious wastes etc.
- **Construction/demolition waste** includes debris and rubbles, wood, concrete etc.
- **Horticulture waste and waste from slaughter houses** include vegetable parts, residues and remains of slaughtered animals, respectively.

The urban solid waste materials that can be degraded by micro-organisms are called **biodegradable wastes**. Examples of this type of waste are vegetable wastes, stale food, tea leaves, egg shells, peanut shells, dry leaves etc. Wastes that cannot be degraded by micro-organisms are called **non-biodegradable wastes**. For example, polyethylene bags, scrap metal, glass bottles etc.

- **Industrial waste:** Industrial waste consists of a large number of materials including factory rubbish, packaging material, organic wastes, acids, alkalis and metals etc. During some industrial processing large quantities of hazardous and toxic materials are also produced. The main sources of industrial wastes are chemical industries, metal and mineral processing industries. Radioactive wastes are generated by nuclear power plants. Thermal power plants produce fly ash in large quantities. Solid wastes from other types of industries include scrap metal, rubber, plastic, paper, glass, wood, oils, paints, asphalt, tars, dyes, scrap leather, ceramics, abrasives, slag, heavy metals, asbestos, batteries. In Europe and North America the environmental laws and safety laws are becoming more stringent due to which disposal of hazardous wastes is becoming a problem. Cost of disposal of such wastes is increasing. Therefore, these wastes are being exported to developing countries which do not even have sufficient knowledge or technique for their disposal.

(B) Effects of Solid Wastes: Municipal solid wastes heap up on the roads due to improper disposal system. People clean their own houses and litter their immediate surroundings which affects the community including themselves. This type of dumping allows biodegradable materials to decompose

under uncontrolled and unhygienic conditions. This produces foul smell and breeds various types of insects and infectious organisms besides spoiling the aesthetics of the site.

Industrial solid wastes are sources of toxic metals and hazardous wastes, which may spread on land and can cause changes in physico-chemical and biological characteristics thereby affecting productivity of soils. Toxic substances may leach or percolate to contaminate the ground water.

In refuse mixing, the hazardous wastes are mixed with garbage and other combustible wastes. This makes segregation and disposal all the more difficult and risky. Various types of wastes like cans, pesticides, cleaning solvents, batteries (zinc, lead or mercury), radioactive materials, plastics and e-waste are mixed up with paper, scraps and other non-toxic materials which could be recycled. Burning of some of these materials produces dioxins, furans and polychlorinated biphenyls, which have the potential to cause various types of ailments including cancer.

(C) Management of Solid Waste: For waste management we stress on 'three R's'-Reduce, Reuse and Recycle before destruction and safe storage of wastes.

(i) Reduction in use of raw materials: Reduction in the use of raw materials will correspondingly decrease the production of waste. Reduced demand for any metallic product will decrease the mining of their metal and cause less production of waste.

(ii) Reuse of waste materials: The refillable containers which are discarded after use can be reused. Villagers make casseroles and silos from waste paper and other waste materials. Making rubber rings from the discarded cycle tubes which are used by the newspaper vendors, instead of rubber bands, reduces the waste generation during manufacturing of rubber bands. Because of financial constraints poor people reuse their materials to the maximum.

(iii) Recycling of materials: Recycling is the reprocessing of discarded materials into new useful products.

- Formation of some old type products *e.g.* old aluminium cans and glass bottles are melted and recast into new cans and bottles.
- Formation of new products: Preparation of cellulose insulation from paper, preparation of fuel pellets from kitchen waste. Preparation of automobiles and construction materials from steel cans.

The process of reducing, reusing and recycling saves money, energy, raw materials, land space and also reduces pollution. Recycling of paper will reduce cutting of trees for making fresh paper. Reuse of metals will reduce mining and melting of ores for recovery of metals from ores and prevent pollution.

For **discarding wastes** the following methods can be adopted:

(i) Sanitary landfill: In a sanitary landfill, garbage is spread out in thin layers, compacted and covered with clay or plastic foam.

In the modern landfills the bottom is covered with an impermeable liner, usually several layers of clay, thick plastic and sand. The liner protects the ground water from being contaminated due to percolation of leachate. Leachate from bottom is pumped and sent for treatment. When landfill is full it is covered with clay, sand, gravel and top soil to prevent seepage of water. Several wells are drilled near the landfill site to monitor if any leakage is contaminating ground water. Methane produced by anaerobic decomposition is collected and burnt to produce electricity or heat.

(ii) **Composting:** Due to shortage of space for landfill in bigger cities, the biodegradable yard waste (kept separate from the municipal waste) is allowed to degrade or decompose in an oxygen rich medium. A good quality nutrient rich and environmental friendly manure is formed which improves the soil conditions and fertility.

(iii) **Incineration:** Incinerators are burning plants capable of burning a large amount of materials at high temperature generally more than 900°C . The required heat comes from oxidation of organically bound carbon and hydrogen present in the waste material or the added fuel. $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + \text{heat}$. The initial cost is very high. During incineration high levels of dioxins, furans, lead and cadmium may be emitted with the fly ash of incinerator. Dioxin level may reach many times more than in the ambient environment. For incineration of materials, it is better to remove batteries containing heavy metals and plastic containing chlorine before burning the material. Prior removal of plastics will reduce emissions of dioxins and polychlorinated biphenyls (PCBs).

I. QUESTIONS

1. Define pollution. Name various atmospheric pollutants.
2. What are the natural and man made pollutants that cause air pollution?
3. Give an account of the adverse effects of air pollution.
4. Differentiate between
 - (i) Sound and noise
 - (ii) Loudness and pitch of noise
 - (iii) Threshold of pain and threshold of hearing for a human ear.
5. Briefly describe the sources, effects and control of noise pollution.
6. Give an account of noise generated during diwali. What would you suggest to reduce this menace ?
7. Enumerate with examples the major sources of surface water pollution and ground water pollution.
8. Write short notes on
 - (a) Minamata disease
 - (b) Biomagnification
 - (c) Itai-itai disease
 - (d) Blue baby syndrome
 - (e) B.O.D.
9. Discuss adverse effects and control of water pollution.
10. What are the major sources of soil pollution ? How does soil pollution affect soil productivity ? What measures can be taken to prevent soil pollution ?
11. Classify solid waste. What are the sources of urban and industrial solid wastes ?
12. What adverse effects can solid wastes cause ? How can the solid waste be managed ?
13. Write short notes on:
 - (a) Donora air pollution episode
 - (b) Bhopal gas tragedy
 - (c) Love canal tragedy
 - (d) Chernobyl nuclear disaster.

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. forms the highest proportion in the vehicular exhaust.
2. Sulphur dioxide during coal burning is produced due to oxidation of contained in coal.
3. CO has affinity for haemoglobin times more than oxygen.
4. Air pollutants affects plants by entering through
5. Sound frequency is expressed in
6. Noise levels considered as threshold of pain are dB.
7. As per Environmental (Protection) (Second Amendment) Rules, 1999 the permissible noise levels for fire-crackers are dB.
8. Minamata disease occurred due to consumption of fish contaminated with
9. Blue baby syndrome is caused by the presence of in drinking water.

(B) TRUE OR FALSE

1. Benzo- α -pyrene of cigarette smoke is considered to cause cancer. (True/False)
2. SO₂ does not affect respiratory passage. (True/False)
3. Washing of coal cannot remove sulphur. (True/False)
4. Sound can propagate without any medium. (True/False)
5. Rocket engine causes 90 dB of noise. (True/False)
6. On Diwali the use of fire-crackers is not permitted between 6.00 pm and 10.00 pm. (True/False)
7. Groundwater is not less prone to contamination due to soil mantle. (True/False)
8. Groundwater contamination is irreversible. (True/False)
9. B.O.D. is always higher than C.O.D. (True/False)
10. High temperature increases the dissolved oxygen content in water. (True/False)
11. Solid waste material degraded by micro-organisms are called non-biodegradable. (True/False)

(C) CHOOSE THE CORRECT ANSWER

1. Damage to leaf structure by air pollutants causes
 - (a) Dead areas of leaf
 - (b) Chlorophyll reduction
 - (c) Dropping of leaf
 - (d) All of these.
2. Air pollutants mixing up with rain can cause
 - (a) High acidity
 - (b) Low acidity
 - (c) Neutral conditions
 - (d) None of these.
3. Industrial wastes may contain toxic
 - (a) Chemicals
 - (b) Phenols
 - (c) Acids
 - (d) All of these.

4. Dissolved oxygen in water comes from
 - (a) Photosynthesis of aquatic plants
 - (b) Atmosphere
 - (c) None of these
 - (d) Both of these.
5. Itai itai disease in Japan was caused by consumption of rice contaminated with
 - (a) Mercury
 - (b) Iron
 - (c) Cadmium
 - (d) Zinc.
6. Oil in water affects fish by affecting
 - (a) Gills
 - (b) Scales
 - (c) Eyes
 - (d) None of these.
7. Which of the following have more penetration power ?
 - (a) Alpha particles
 - (b) Beta particles
 - (c) Gamma-rays
 - (d) None of these.
8. Bhopal gas tragedy occurred due to leakage of
 - (a) MIC
 - (b) DDT
 - (c) SO₂
 - (d) Dioxins.